Erasmus University Rotterdam
Erasmus School of Economics
Master's Thesis Econometrics and Management Science - Business Analytics and Quantitative Marketing

Median-of-Means Two-Stage Least Squares Regression

David von Schledorn Márquez (597334)

Supervisor:	Mikhail Zhelonkin
Second assessor:	-
Date final version:	-

Abstract

Median-of-Means methods are used for instrumental variable regression and their properties are compared to the standard 2-stage least squares method.

1 Preliminaries

1.1 Setup and Assumptions

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample from a joint distribution on $\mathbb{R}^3$ satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i, \tag{1}$$

where $\beta \in \mathbb{R}$ is the structural parameter of interest and ε_i is an unobserved error term. We impose the following conditions.

(A1) **Exogeneity.** $\mathbb{E}[Z_i \varepsilon_i] = 0$.

(A2) **Relevance.** $\mu_{ZX} := \mathbb{E}[Z_i X_i] \neq 0$.

(A3) **Finite second moments.** $\sigma_{ZY}^2 := \text{Var}(Z_i Y_i) < \infty$, $\sigma_{ZX}^2 := \text{Var}(Z_i X_i) < \infty$, and $\sigma_{Z\varepsilon}^2 := \text{Var}(Z_i \varepsilon_i) < \infty$.

Define $\mu_{ZY} := \mathbb{E}[Z_i Y_i]$. By (A1) and (A2), the structural parameter is identified as $\beta = \mu_{ZY}/\mu_{ZX}$.

Make clearer

Remark 1.1 (Moment conditions). *The conditions (A1)–(A3) hold for any distribution with finite second moments of the products $Z_i Y_i$, $Z_i X_i$, and $Z_i \varepsilon_i$. Thus, no assumptions are imposed on the marginal distribiutions of Y_i, X_i, Z_i, or ε_i.*

Remark 1.2 (Relationship between variances). *The three variances in (A3) are related. Since $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$, Minkowski's inequality gives*

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta|\, \sigma_{ZX}. \tag{2}$$

To see this, set $U = Z_i Y_i$ and $V = \beta Z_i X_i$, so that $U - V = Z_i \varepsilon_i$. By Minkowski's inequality for $p = 2$,

$$\sqrt{\text{Var}(U - V)} \leq \sqrt{\text{Var}(U)} + \sqrt{\text{Var}(V)},$$

which gives $\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta|\, \sigma_{ZX}$.

1.2 Sub-Gaussian Estimators

Definition 1.3 (Sub-Gaussian Estimators). *Let $(X_i)_{i=1}^n$ be a random sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, and let $\bar{\mu}(X)$ denote an estimator for μ. For some constant $L > 0$, $\bar{\mu}(X)$ is said to be a sub-Gaussian estimator for μ if*

$$\mathbb{P}\left[|\bar{\mu}(X) - \mu| > L\sigma \sqrt{\frac{\ln(1/\delta)}{n}} \right] \leq \delta \quad \textit{for all } \delta \in (0,1). \tag{3}$$

1.3 The Median-of-Means Estimator for a Scalar Mean

We present the scalar Median-of-Means (MoM) estimator and its concentration properties. The results in this section are adapted from Devroye, Lerasle, Lugosi and Oliveira (2016) and Lugosi and Mendelson (2019).

Algorithm 1: Scalar Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\mu}$

Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \ldots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \ldots, k$ **do**

  Compute the block mean $\bar{\mu}_j = \frac{1}{m} \sum_{i \in B_j} X_i$;

return $\hat{\mu} = \mathrm{med}(\bar{\mu}_1, \ldots, \bar{\mu}_k)$;

Theorem 1.4 (MoM concentration). *Let $(X_i)_{i=1}^n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Let k be a positive integer, $m = \lfloor n/k \rfloor$, and $\hat{\mu}_{\mathrm{MoM}}$ the MoM estimator with k blocks. Then*

$$\mathbb{P}\left[|\hat{\mu}_{\mathrm{MoM}} - \mu| > \frac{2\sigma}{\sqrt{m}}\right] \leq e^{-k/8}. \tag{4}$$

Setting $k = \lceil 8 \ln(1/\delta) \rceil$ for $\delta \in (0,1)$, this becomes

$$\mathbb{P}\left[|\hat{\mu}_{\mathrm{MoM}} - \mu| > \sigma \sqrt{\frac{32 \ln(1/\delta)}{n}}\right] \leq \delta, \tag{5}$$

which is sub-Gaussian in the sense that the dependence on δ is logarithmic.

Proof. The proof proceeds in three steps.

Step 1 (Chebyshev on each block). Each block mean satisfies $\mathbb{E}[\bar{X}_j] = \mu$ and $\mathrm{Var}(\bar{X}_j) = \sigma^2/m$. By Chebyshev's inequality with threshold $2\sigma/\sqrt{m}$,

$$\mathbb{P}\left[|\bar{X}_j - \mu| > \frac{2\sigma}{\sqrt{m}}\right] \leq \frac{\sigma^2/m}{4\sigma^2/m} = \frac{1}{4}.$$

Step 2 (Counting argument for the median). Define $\zeta_j = \mathbf{1}\{|\bar{X}_j - \mu| > 2\sigma/\sqrt{m}\}$. By Step 1, $\mathbb{E}[\zeta_j] \leq 1/4$. Since the blocks are disjoint subsets of an i.i.d. sample, the $(\zeta_j)_{j=1}^k$ are independent.

If $|\hat{\mu}_{\mathrm{MoM}} - \mu| > 2\sigma/\sqrt{m}$, the median exceeds $\mu + 2\sigma/\sqrt{m}$ or falls below $\mu - 2\sigma/\sqrt{m}$. In either case, at least $k/2$ of the block means satisfy $|\bar{X}_j - \mu| > 2\sigma/\sqrt{m}$, since the median of k numbers can only lie outside an interval if fewer than half the numbers lie inside it. Therefore,

$$\mathbb{P}\left[|\hat{\mu}_{\mathrm{MoM}} - \mu| > \frac{2\sigma}{\sqrt{m}}\right] \leq \mathbb{P}\left[\sum_{j=1}^{k} \zeta_j \geq \frac{k}{2}\right].$$

Step 3 (Hoeffding's inequality). The sum $\sum_{j=1}^k \zeta_j$ consists of k independent random variables with $\zeta_j \in [0,1]$ and $\mathbb{E}[\sum_j \zeta_j] \leq k/4$. For $\sum_j \zeta_j$ to reach $k/2$, it must exceed its mean by at least $k/2 - k/4 = k/4$. By Hoeffding's inequality (Hoeffding, 1963),

$$\mathbb{P}\left[\sum_{j=1}^{k} \zeta_j - \mathbb{E}\left[\sum_j \zeta_j\right] \geq \frac{k}{4}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

This establishes (4). The sub-Gaussian form (5) follows by setting $k = \lceil 8 \ln(1/\delta) \rceil$ and noting $m \geq n/(2k)$ for $n \geq 2k$. $\square$

2 The Standard Instrumental Variable Estimator

Definition 2.1 (Standard IV estimator)**.** *Consider the structural equation* (1)*. The standard IV estimator* $\hat{\beta}_{\mathrm{IV}}$ *is defined by*

$$\hat{\beta}_{\mathrm{IV}} = \frac{\bar{S}_{ZY}}{\bar{S}_{ZX}}, \qquad \bar{S}_{ZY} = \frac{1}{n}\sum_{i=1}^{n} Z_i Y_i, \quad \bar{S}_{ZX} = \frac{1}{n}\sum_{i=1}^{n} Z_i X_i.$$

Lemma 2.2 (Error decomposition)**.** *Under* (1) *and (A1),*

$$\hat{\beta}_{\mathrm{IV}} - \beta = \frac{\bar{S}_{Z\varepsilon}}{\bar{S}_{ZX}}, \tag{6}$$

where $\bar{S}_{Z\varepsilon} = \frac{1}{n}\sum_{i=1}^{n} Z_i \varepsilon_i$ *satisfies* $\mathbb{E}[\bar{S}_{Z\varepsilon}] = 0$.

Proof. Multiply both sides of $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average:

$$\bar{S}_{ZY} = \frac{1}{n}\sum_{i=1}^{n} Z_i(\beta X_i + \varepsilon_i) = \beta \bar{S}_{ZX} + \bar{S}_{Z\varepsilon}.$$

Rearranging and dividing by $\bar{S}_{ZX}$ gives $\hat{\beta}_{\mathrm{IV}} = \beta + \bar{S}_{Z\varepsilon}/\bar{S}_{ZX}$, which is (6). By (A1), $\mathbb{E}[Z_i \varepsilon_i] = 0$, so $\mathbb{E}[\bar{S}_{Z\varepsilon}] = 0$ by linearity of expectation. $\square$

Theorem 2.3 (Standard IV concentration)**.** *Assume (A1)–(A3). For any* $\delta \in (0, 1)$*, if*

$$n \geq \frac{8\sigma_{ZX}^2}{\delta \mu_{ZX}^2}, \tag{7}$$

then

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}\right] \leq \delta. \tag{8}$$

Proof. By Lemma 2.2, the estimation error equals the ratio $\bar{S}_{Z\varepsilon}/\bar{S}_{ZX}$. To bound this ratio we need two things simultaneously: the denominator must be bounded away from zero, and the numerator must be small. We control each requirement with a separate probability event and then combine them via the union bound.

Step 1 (Two-event decomposition). For a denominator threshold $D \in (0, |\mu_{ZX}|)$ and an error threshold $t > 0$, both to be chosen below, define

$$A(D) = \left\{\left|\bar{S}_{ZX}\right| \geq D\right\}, \qquad B(D, t) = \left\{\left|\bar{S}_{Z\varepsilon}\right| \leq tD\right\}.$$

On $A(D) \cap B(D,t)$, the denominator satisfies $\left|\bar{S}_{ZX}\right| \geq D$ by definition of $A(D)$, so

$$\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| = \frac{\left|\bar{S}_{Z\varepsilon}\right|}{\left|\bar{S}_{ZX}\right|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t\right] \leq \mathbb{P}[A(D)^c] + \mathbb{P}[B(D,t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$, the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). We bound each failure probability in turn.

The failure event $A^c = \{\left|\bar{S}_{ZX}\right| < |\mu_{ZX}|/2\}$ implies a large deviation of $\bar{S}_{ZX}$ from its mean. Specifically, if $\left|\bar{S}_{ZX}\right| < |\mu_{ZX}|/2$ then, by the reverse triangle inequality,

$$\left|\bar{S}_{ZX} - \mu_{ZX}\right| \geq |\mu_{ZX}| - \left|\bar{S}_{ZX}\right| > |\mu_{ZX}| - \frac{|\mu_{ZX}|}{2} = \frac{|\mu_{ZX}|}{2}.$$

Hence $A^c \subseteq \{\left|\bar{S}_{ZX} - \mu_{ZX}\right| > |\mu_{ZX}|/2\}$. Since $\mathrm{Var}(\bar{S}_{ZX}) = \sigma^2_{ZX}/n$, Chebyshev's inequality applied with threshold $|\mu_{ZX}|/2$ gives

$$\mathbb{P}[A^c] \leq \frac{\sigma^2_{ZX}/n}{\mu^2_{ZX}/4} = \frac{4\sigma^2_{ZX}}{n\mu^2_{ZX}}.$$

The failure event is $B^c = \{\left|\bar{S}_{Z\varepsilon}\right| > t\,|\mu_{ZX}|/2\}$. Since $\mathbb{E}[\bar{S}_{Z\varepsilon}] = 0$ and $\mathrm{Var}(\bar{S}_{Z\varepsilon}) = \sigma^2_{Z\varepsilon}/n$, Chebyshev's inequality gives

$$\mathbb{P}[B^c] \leq \frac{\sigma^2_{Z\varepsilon}/n}{t^2\mu^2_{ZX}/4} = \frac{4\sigma^2_{Z\varepsilon}}{nt^2\mu^2_{ZX}}.$$

Step 3 (Allocate the error budget and solve for t). We require the total failure probability $\mathbb{P}[A^c] + \mathbb{P}[B^c]$ to be at most δ, and we allocate half the budget to each event.

Budget for A^c. Setting $\mathbb{P}[A^c] \leq \delta/2$ requires

$$\frac{4\sigma^2_{ZX}}{n\mu^2_{ZX}} \leq \frac{\delta}{2} \iff n \geq \frac{8\sigma^2_{ZX}}{\delta\mu^2_{ZX}},$$

which is exactly condition (7).

Setting $\mathbb{P}[B^c] \leq \delta/2$ and solving for t:

$$\frac{4\sigma^2_{Z\varepsilon}}{nt^2\mu^2_{ZX}} = \frac{\delta}{2} \implies t = \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}.$$

By the union bound,

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta. \qquad \square$$

Remark 2.4 (Polynomial dependence on δ)**.** *The bound* (8) *contains the factor* $1/\sqrt{\delta}$*, which is polynomial in* $1/\delta$*. Devroye et al. (2016) show that this is unavoidable: for any estimator formed from the empirical mean of i.i.d. observations with only finite variance, the* $1/\sqrt{\delta}$ *rate cannot be improved without additional distributional assumptions. The MoM-based estimators* *in Sections 3–4 circumvent this by operating on block means.*

Remark 2.5 (The symmetric threshold is not optimal)**.** *The choice $D = |\mu_{ZX}|/2$ was made for simplicity. It splits the distance between* 0 *and $|\mu_{ZX}|$ evenly, but this is not the split that minimises the error bound for given n and δ. Section 2.1 shows that treating D as a free parameter and optimising it reduces the instrument strength constant from* 8 *to* 1 *and improves the error constant by a factor of $2\sqrt{2}$.*

2.1 Optimised Threshold

The proof of Theorem 2.3 introduced a general threshold D but then fixed it symetrically. Here we keep D free and choose it to minimise the resulting error bound symmetrically, at the cost of a more involved Step 3.

Theorem 2.6 (Standard IV concentration, optimised threshold)**.** *Assume (A1)–(A3). Define*

$$\rho = \frac{\sigma_{ZX}^2}{\delta n \mu_{ZX}^2}.$$

For any $\delta \in (0, 1)$, if $\rho < 1$ (i.e., $n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$), then the choice $D^* = (1 - \rho^{1/3})\,|\mu_{ZX}|$ is optimal and

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1 - \rho^{1/3})^{3/2}}\right] \le \delta. \tag{9}$$

Proof. Steps 1 and 2 are identical to those of Theorem 2.3 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A(D)^c] \le \frac{\sigma_{ZX}^2/n}{(|\mu_{ZX}| - D)^2}, \qquad \mathbb{P}[B(D,t)^c] \le \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Requiring their sum to be at most δ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{n(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2 D^2} \le \delta. \tag{10}$$

Step 3 (Optimise D, then solve for t). Write $D = \eta\,|\mu_{ZX}|$ with $\eta \in (0, 1)$. Dividing (20) through by δ and substituting $\rho = \sigma_{ZX}^2/(\delta n \mu_{ZX}^2)$:

$$\frac{\rho}{(1-\eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2 \eta^2 t^2} \le 1.$$

Solving for t^2:

$$t^2 \ge \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2} \cdot \frac{1}{f(\eta)}, \qquad f(\eta) = \eta^2\left(1 - \frac{\rho}{(1-\eta)^2}\right).$$

Minimising t is equivalent to maximising f over $\eta \in (0, 1)$. Differentiating and setting $f'(\eta) = 0$:

$$f'(\eta) = 2\eta\left(1 - \frac{\rho}{(1-\eta)^2}\right) - \frac{2\rho\eta^2}{(1-\eta)^3} = 0.$$

Dividing through by $2\eta > 0$:

$$1 - \frac{\rho}{(1-\eta)^2} - \frac{\rho\eta}{(1-\eta)^3} = 0 \iff (1-\eta)^3 - \rho(1-\eta) - \rho\eta = 0 \iff (1-\eta)^3 = \rho.$$

Hence $\eta^* = 1 - \rho^{1/3}$, which lies in $(0,1)$ if and only if $\rho < 1$, the stated instrument strength condition. Substituting $(1-\eta^*)^2 = \rho^{2/3}$:

$$f(\eta^*) = (1-\rho^{1/3})^2\left(1 - \frac{\rho}{\rho^{2/3}}\right) = (1-\rho^{1/3})^2(1-\rho^{1/3}) = (1-\rho^{1/3})^3.$$

The minimised threshold is therefore

$$t^* = \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1-\rho^{1/3})^{3/2}},$$

and the union bound gives $\mathbb{P}[\left|\hat{\beta}_{IV} - \beta\right| > t^*] \leq \delta$. $\square$

Remark 2.7 (Comparison with the symmetric bound)**.** *As $\rho \to 0$ (strong instruments or large n), the correction factor $(1-\rho^{1/3})^{-3/2} \to 1$ and (19) approaches $\sigma_{Z\varepsilon}/(|\mu_{ZX}|\sqrt{\delta n})$, a factor of $2\sqrt{2} \approx 2.83$ tighter than (8). As $\rho \to 1$ (approaching the boundary of instrument relevance), the factor diverges, reflecting the genuine difficulty of ratio estimation near instrument irrelevance. The instrument strength condition is eight times weaker than Theorem 2.3: $\rho < 1$ versus $\rho < 1/8$.*

2.2 Cantelli Improvement

The Chebyshev bound applied to $A(D)^c$ in the proofs above is two-sided. However, $A(D)^c = \{|\bar{S}_{ZX}| < D\}$ is a one-sided failure event: only a *downward* deviation of $\bar{S}_{ZX}$ from μ_{ZX} causes the denominator to be too small; an upward deviation makes it larger and is harmless. Cantelli's inequality exploits this one-sidedness.

Lemma 2.8 (Cantelli's inequality)**.** *Let W be a random variable with $\mathbb{E}[W] = \mu$ and $\mathrm{Var}(W) = \sigma^2 < \infty$. For any $\tau > 0$,*

$$\mathbb{P}[W - \mu \leq -\tau] \leq \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

This is strictly smaller than the Chebyshev bound σ^2/τ^2 for all $\tau > 0$, with the greatest relative gain when $\tau \approx \sigma$.

Theorem 2.9 (Standard IV concentration, Cantelli improvement)**.** *Assume (A1)–(A3) and $\mu_{ZX} > 0$. Write $\gamma_{\mathrm{IV}} = \sigma^2_{ZX}/(n\mu^2_{ZX})$ and define*

$$h(\eta) = \eta^2 \frac{\delta\big(\gamma_{\mathrm{IV}} + (1-\eta)^2\big) - \gamma_{\mathrm{IV}}}{\gamma_{\mathrm{IV}} + (1-\eta)^2}, \quad \eta \in (0,1). \tag{11}$$

If

$$n > \frac{(1-\delta)\sigma^2_{ZX}}{\delta\mu^2_{ZX}}, \quad \textit{equivalently,} \quad \gamma_{\mathrm{IV}} < \frac{\delta}{1-\delta}, \tag{12}$$

then $h(\eta) > 0$ on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{nh(\eta^*_{\mathrm{IV}})}}\right] \leq \delta, \tag{13}$$

where η^*_{IV} is the maximiser of h over its feasible domain.

Proof. We use the same two-event framework with $D = \eta\mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). As in Theorem 2.3, define $A(D) = \{\bar{S}_{ZX} \geq D\}$ and $B(D,t) = \{\left|\bar{S}_{Z\varepsilon}\right| \leq tD\}$. On $A(D) \cap B(D,t)$, $\left|\hat{\beta}_{IV} - \beta\right| \leq t$.

Step 2 (Cantelli on A^c, Chebyshev on B^c).

Since $\mu_{ZX} > 0$ and $D < \mu_{ZX}$, the failure event satisfies

$$A(D)^c = \{\bar{S}_{ZX} < D\} \subseteq \{\bar{S}_{ZX} - \mu_{ZX} < -(\mu_{ZX} - D)\}.$$

This is a one-sided downward deviation, so Lemma 4.7 applies with $W = \bar{S}_{ZX}$, $\sigma^2 = \sigma^2_{ZX}/n$, and $\tau = \mu_{ZX} - D$:

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma^2_{ZX}/n}{\sigma^2_{ZX}/n + (\mu_{ZX} - D)^2}.$$

The event $B(D,t)^c = \{\left|\bar{S}_{Z\varepsilon}\right| > tD\}$ is two-sided (a large residual of either sign causes failure), so Chebyshev is appropriate:

$$\mathbb{P}[B(D,t)^c] \leq \frac{\sigma^2_{Z\varepsilon}/n}{t^2D^2}.$$

Step 3 (Constraint and optimization). Setting the total failure probability to at most δ and substituting $D = \eta\mu_{ZX}$, $\gamma_{\mathrm{IV}} = \sigma^2_{ZX}/(n\mu^2_{ZX})$:

$$\frac{\gamma_{\mathrm{IV}}}{\gamma_{\mathrm{IV}} + (1-\eta)^2} + \frac{\sigma^2_{Z\varepsilon}}{nt^2\eta^2\mu^2_{ZX}} \leq \delta.$$

Isolating the second term and solving for t^2:

$$t^2 \geq \frac{\sigma^2_{Z\varepsilon}}{n\mu^2_{ZX}h(\eta)},$$

where $h(\eta)$ is defined in (21). The function $h(\eta)$ is positive only when $\delta(\gamma_{\mathrm{IV}} + (1-\eta)^2) > \gamma_{\mathrm{IV}}$, i.e., $(1-\eta)^2 > \gamma_{\mathrm{IV}}(1-\delta)/\delta$. The feasible region is non-empty if and only if $\gamma_{\mathrm{IV}} < \delta/(1-\delta)$, which is condition (22). Minimizing t over the feasible region by maximizing h and applying the union bound gives (23). □

Remark 2.10 (When Cantelli helps). *Compared with the optimised Chebyshev condition ($n > \sigma^2_{ZX}/(\delta\mu^2_{ZX})$), the Cantelli condition* (22) *is weaker by a factor of* $1/(1-\delta)$*. At* $\delta = 0.05$ *this ratio is* $1/0.95 \approx 1.05$*, so the gain is negligible for typical significance levels. The improvement is more meaningful at moderate confidence levels: at* $\delta = 0.5$ *the factor reaches* 2*. For the standard IV estimator the Cantelli improvement is therefore primarily of theoretical interest. By contrast, for the MoR estimator (Section* **??***), Cantelli extends the valid range of the instrument strength parameter from* $\gamma < 1/4$ *to* $\gamma < 1/3$*, a qualitatively more substantial gain.*

Remark 2.11 (No closed form for η^*_{IV})**.** *The first-order condition $h'(\eta) = 0$ reduces to a quartic in η with no general closed-form solution. The optimised Chebyshev version (Theorem 4.5) yielded the clean form $\eta^* = 1 - \rho^{1/3}$ because the Chebyshev bound σ^2/τ^2 produces a simpler algebraic structure than the Cantelli bound $\sigma^2/(\sigma^2 + \tau^2)$. The bound (23) must therefore be evaluated numerically for specific parameter values.*

3 The Ratio-of-Medians MoM-IV Estimator

Algorithm 2: Ratio-of-Medians Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k
Output: Estimate $\hat{\beta}_{\mathrm{RM}}$
Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \ldots, B_k$ of size $m = \lfloor n/k \rfloor$;
for $j = 1, \ldots, k$ **do**
  Compute the block means $\hat{S}_{ZY}^{(j)} = \frac{1}{m}\sum_{i \in B_j} Z_i Y_i$ and $\hat{S}_{ZX}^{(j)} = \frac{1}{m}\sum_{i \in B_j} Z_i X_i$;
  Compute coordinate-wise medians $\widetilde{S}_{ZY} = \mathrm{med}\left(\hat{S}_{ZY}^{(1)}, \ldots, \hat{S}_{ZY}^{(k)}\right)$ and
  $\widetilde{S}_{ZX} = \mathrm{med}\left(\hat{S}_{ZX}^{(1)}, \ldots, \hat{S}_{ZX}^{(k)}\right)$;
return $\hat{\beta}_{\mathrm{RM}} = \frac{\widetilde{S}_{ZY}}{\widetilde{S}_{ZX}}$;

Fix formatting

Theorem 3.1 (RoM concentration)**.** *Assume (A1)–(A3). Let $\delta \in (0,1)$, $k = \lceil 8\ln(2/\delta)\rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition*

$$m > \frac{4\sigma_{ZX}^2}{\mu_{ZX}^2} \tag{14}$$

holds, then

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{RM}} - \beta\right| > \frac{2\left(\sigma_{ZY} + |\beta|\,\sigma_{ZX}\right)}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}\right] \le \delta. \tag{15}$$

Proof. The proof applies the scalar MoM result (Theorem 1.4) to each coordinate separately and combines the results with a union bound.

Step 1 (MoM on each coordinate). By Theorem 1.4 applied to the sequences $(Z_iY_i)_{i=1}^n$ and $(Z_iX_i)_{i=1}^n$,

$$\mathbb{P}\left[\left|\widetilde{S}_{ZY} - \mu_{ZY}\right| > \frac{2\sigma_{ZY}}{\sqrt{m}}\right] \le e^{-k/8}, \qquad \mathbb{P}\left[\left|\widetilde{S}_{ZX} - \mu_{ZX}\right| > \frac{2\sigma_{ZX}}{\sqrt{m}}\right] \le e^{-k/8}.$$

Step 2 (Union bound). Define the events

$$A = \left\{\left|\widetilde{S}_{ZY} - \mu_{ZY}\right| \le \frac{2\sigma_{ZY}}{\sqrt{m}}\right\}, \qquad B = \left\{\left|\widetilde{S}_{ZX} - \mu_{ZX}\right| \le \frac{2\sigma_{ZX}}{\sqrt{m}}\right\}.$$

By Step 1, $\mathbb{P}[A^c] \le e^{-k/8}$ and $\mathbb{P}[B^c] \le e^{-k/8}$. Note that A and B are not independent (both are constructed from the same data), so we use the union bound:

$$\mathbb{P}[(A \cap B)^c] = \mathbb{P}[A^c \cup B^c] \le \mathbb{P}[A^c] + \mathbb{P}[B^c] \le 2e^{-k/8}.$$

Setting $k = \lceil 8\ln(2/\delta)\rceil$ ensures $2e^{-k/8} \le \delta$.

Step 3 (Algebraic bound on the ratio). On $A \cap B$, we decompose

$$\hat{\beta}_{\mathrm{RM}} - \beta = \frac{\widetilde{S}_{ZY}}{\widetilde{S}_{ZX}} - \frac{\mu_{ZY}}{\mu_{ZX}} = \frac{\widetilde{S}_{ZY}\mu_{ZX} - \mu_{ZY}\widetilde{S}_{ZX}}{\widetilde{S}_{ZX} \cdot \mu_{ZX}}.$$

Adding and subtracting $\mu_{ZY}\mu_{ZX}$ in the numerator and using $\mu_{ZY} = \beta\mu_{ZX}$,

$$\widetilde{S}_{ZY}\mu_{ZX} - \mu_{ZY}\widetilde{S}_{ZX} = \mu_{ZX}\Big[(\widetilde{S}_{ZY} - \mu_{ZY}) - \beta(\widetilde{S}_{ZX} - \mu_{ZX})\Big].$$

The factor μ_{ZX} cancels with the μ_{ZX} in the denominator, giving

$$\hat{\beta}_{\mathrm{RM}} - \beta = \frac{(\widetilde{S}_{ZY} - \mu_{ZY}) - \beta(\widetilde{S}_{ZX} - \mu_{ZX})}{\widetilde{S}_{ZX}}.$$

Taking absolute values and applying the triangle inequality to the numerator,

$$\left|\hat{\beta}_{\mathrm{RM}} - \beta\right| \le \frac{\left|\widetilde{S}_{ZY} - \mu_{ZY}\right| + |\beta|\left|\widetilde{S}_{ZX} - \mu_{ZX}\right|}{\left|\widetilde{S}_{ZX}\right|}.$$

Step 4 (Substitute bounds). On event A, $\left|\widetilde{S}_{ZY} - \mu_{ZY}\right| \le 2\sigma_{ZY}/\sqrt{m}$. On event B, $\left|\widetilde{S}_{ZX} - \mu_{ZX}\right| \le 2\sigma_{ZX}/\sqrt{m}$, and by the reverse triangle inequality,

$$\left|\widetilde{S}_{ZX}\right| \ge |\mu_{ZX}| - \left|\widetilde{S}_{ZX} - \mu_{ZX}\right| \ge |\mu_{ZX}| - \frac{2\sigma_{ZX}}{\sqrt{m}}.$$

The instrument strength condition (14) ensures $2\sigma_{ZX}/\sqrt{m} < |\mu_{ZX}|$, so the denominator is strictly positive. Substituting,

$$\left|\hat{\beta}_{\mathrm{RM}} - \beta\right| \le \frac{2\sigma_{ZY}/\sqrt{m} + |\beta| \cdot 2\sigma_{ZX}/\sqrt{m}}{|\mu_{ZX}| - 2\sigma_{ZX}/\sqrt{m}} = \frac{2(\sigma_{ZY} + |\beta|\,\sigma_{ZX})}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}.$$

Since this holds on $A \cap B$ and $\mathbb{P}[(A \cap B)^c] \le \delta$, the result follows. □

4 The Median-of-Ratios MoM-IV Estimator

Algorithm 3: Median-of-Ratios Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{\mathrm{RM}}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \ldots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \ldots, k$ **do**

  Compute the block means $\hat{S}_{ZY}^{(j)} = \frac{1}{m}\sum_{i\in B_j} Z_i Y_i$ and $\hat{S}_{ZX}^{(j)} = \frac{1}{m}\sum_{i\in B_j} Z_i X_i$;

  Compute block level IV estimates $\hat{\beta}^{(j)} = \frac{\hat{S}_{ZY}^{(j)}}{\hat{S}_{ZX}^{(j)}}$;

return $\hat{\beta}_{\mathrm{MR}} = \mathrm{median}(\hat{\beta}^{(1)}, \ldots, \hat{\beta}^{(k)})$;

Lemma 4.1 (Block-level error decomposition)**.** *Under* (1) *and (A1), for each block* j*,*

$$\hat{\beta}^{(j)} - \beta = \frac{\bar{\varepsilon}_j^Z}{\hat{S}_{ZX}^{(j)}}, \tag{16}$$

where $\bar{\varepsilon}_j^Z = \frac{1}{m}\sum_{i \in B_j} Z_i \varepsilon_i$ *satisfies* $\mathbb{E}[\bar{\varepsilon}_j^Z] = 0$.

Proof. Multiply $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average over block B_j:

$$\hat{S}_{ZY}^{(j)} = \frac{1}{m}\sum_{i \in B_j} Z_i(\beta X_i + \varepsilon_i) = \beta \hat{S}_{ZX}^{(j)} + \bar{\varepsilon}_j^Z.$$

Dividing by $\hat{S}_{ZX}^{(j)}$ gives $\hat{\beta}^{(j)} = \beta + \bar{\varepsilon}_j^Z / \hat{S}_{ZX}^{(j)}$. By (A1) and linearity of expectation, $\mathbb{E}[\bar{\varepsilon}_j^Z] = \frac{1}{m}\sum_{i \in B_j} \mathbb{E}[Z_i \varepsilon_i] = 0$. □

Theorem 4.2 (MoR concentration)**.** *Assume (A1)–(A3). Let* $\delta \in (0, 1)$*,* $k = \lceil 8 \ln(1/\delta) \rceil$*, and* $m = \lfloor n/k \rfloor$*. If the instrument strength condition*

$$m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2} \tag{17}$$

holds, then

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{MR}} - \beta\right| > \frac{4\sqrt{2}\,\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}}\right] \leq \delta. \tag{18}$$

Proof. The proof applies the MoM counting argument directly to the block-level IV estimates $\hat{\beta}^{(j)}$.

Step 1 (Two-event decomposition). By Lemma 4.1, $\left|\hat{\beta}^{(j)} - \beta\right| = \left|\bar{\varepsilon}_j^Z\right| / \left|\hat{S}_{ZX}^{(j)}\right|$. To bound this ratio, we control the numerator above and the denominator below. For a threshold $t > 0$ to be determined, define

$$A_j(D) = \left\{\left|\hat{S}_{ZX}^{(j)} - \mu_{ZX}\right| \leq D\right\}, \qquad B_j(D, T) = \left\{\left|\bar{\varepsilon}_Z^{(j)}\right| \leq tD\right\}.$$

On $A_j(D) \cap B_j(D, t)$, the denominator satisfies $\left|\bar{S}_{ZX}\right| \geq D$ by definition of $A_j(D)$, so

$$\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| = \frac{\left|\bar{\varepsilon}_Z^{(j)}\right|}{\left|\bar{S}_{ZX}\right|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P}\left[\left|\hat{\beta}_{MR} - \beta\right| > t\right] \leq \mathbb{P}[A_j(D)^c] + \mathbb{P}[B_j(D,t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$, the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). We bound each failure probability in turn.

The failure event $A_j^c = \{\left|\bar{S}_{ZX}^{(j)}\right| < |\mu_{ZX}|/2\}$ implies a large deviation of $\bar{S}_{ZX}^{(j)}$ from its mean. Specifically, if $\left|\bar{S}_{ZX}^{(j)}\right| < |\mu_{ZX}|/2$ then, by the reverse triangle inequality,

$$\left|\bar{S}_{ZX}^{(j)} - \mu_{ZX}\right| \geq |\mu_{ZX}| - \left|\bar{S}_{ZX}^{(j)}\right| > |\mu_{ZX}| - \frac{|\mu_{ZX}|}{2} = \frac{|\mu_{ZX}|}{2}.$$

Hence $A_j^c \subseteq \{\left|\bar{S}_{ZX}^{(j)} - \mu_{ZX}\right| > |\mu_{ZX}|/2\}$. Since $\mathrm{Var}(\bar{S}_{ZX}^{(j)}) = \sigma_{ZX}^2/m$, Chebyshev's inequality applied with threshold $|\mu_{ZX}|/2$ gives

$$\mathbb{P}[A_j^c] \leq \frac{\sigma_{ZX}^2/m}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

The failure event is $B_j^c = \{\left|\bar{\varepsilon}_Z^{(j)}\right| > t\,|\mu_{ZX}|/2\}$. Since $\mathbb{E}[\bar{\varepsilon}_Z^{(j)}] = 0$ and $\mathrm{Var}(\varepsilon_Z^{\bar{(j)}}) = \sigma_{Z\varepsilon}^2/m$, Chebyshev's inequality gives

$$\mathbb{P}[B_j^c] \leq \frac{\sigma_{Z\varepsilon}^2/m}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{mt^2\mu_{ZX}^2}.$$

The instrument strength condition (17) ensures $\mathbb{P}[A_j^c] \leq 1/8$. Setting $t = 4\sqrt{2}\,\sigma_{Z\varepsilon}/(|\mu_{ZX}|\sqrt{m})$ gives $\mathbb{P}[B_j^c] \leq 1/8$.

Step 3 (Allocate the error budget and solve for t). We require the total failure probability $\mathbb{P}[A^c] + \mathbb{P}[B^c]$ to be at most $\frac{1}{4}$, and we allocate half the budget to each event.

Budget for A_j^c. Setting $\mathbb{P}[A^c] \leq 1/8$ requires

$$\frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2} \leq \frac{1}{8} \iff m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2},$$

which is exactly condition (17).

Setting $\mathbb{P}[B_j^c] \leq 1/8$ and solving for t:

$$\frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2} \leq \frac{1}{8} \implies t \geq \frac{\sqrt{32}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}}.$$

By the union bound,

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{8}.$$

Step 4 (Counting argument). Define $\zeta_j = \mathbf{1}\{|\hat{\beta}^{(j)} - \beta| > t\}$. By Step 2, $\mathbb{E}[\zeta_j] \leq 1/4$. Since the blocks are disjoint, the $(\zeta_j)_{j=1}^k$ are independent. If $|\hat{\beta}_{\mathrm{MR}} - \beta| > t$, the median of the block estimates lies outside $(\beta - t, \beta + t)$, which requires at least $k/2$ blocks to satisfy $|\hat{\beta}^{(j)} - \beta| > t$. Therefore,

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{MR}} - \beta\right| > t\right] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right].$$

Step 5 (Hoeffding). By Hoeffding's inequality, with overshoot $k/2 - k/4 = k/4$,

$$\mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

Setting $k = \lceil 8\ln(1/\delta)\rceil$ gives $e^{-k/8} \leq \delta$, completing the proof. $\square$

Remark 4.3 (Logarithmic dependence on δ)**.** *The bound* (18) *contains the factor* $\ln(1/\delta)$*, which is polynomial in* $1/\delta$*.*

Remark 4.4 (The symmetric threshold is not optimal)**.** *The choice* $D = |\mu_{ZX}|/2$ *was made for simplicity. It splits the distance between* 0 *and* $|\mu_{ZX}|$ *evenly, but this is not the split that minimises the error bound for given* n *and* δ*. Section 4.1 shows that treating* D *as a free parameter and optimising it reduces the instrument strength constant from something to something else and improves the error constant by a factor of another thing.*

add the stuff

keep going

4.1 Optimised Threshold

The proof of Theorem 4.2 introduced a general threshold D but then fixed it symetrically. Here we keep D free and choose it to minimise the resulting error bound symmetrically, at the cost of a more involved Step 3.

Theorem 4.5 (Standard IV concentration, optimised threshold)**.** *Assume (A1)–(A3). Define*

$$\rho = \frac{\sigma_{ZX}^2}{\delta n \mu_{ZX}^2}.$$

For any $\delta \in (0,1)$*, if* $\rho < 1$ *(i.e.,* $n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$*), then the choice* $D^* = (1-\rho^{1/3})\,|\mu_{ZX}|$ *is optimal and*

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1-\rho^{1/3})^{3/2}}\right] \le \delta. \tag{19}$$

Proof. Steps 1 and 2 are identical to those of Theorem 2.3 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A(D)^c] \le \frac{\sigma_{ZX}^2/n}{(|\mu_{ZX}| - D)^2}, \qquad \mathbb{P}[B(D,t)^c] \le \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Requiring their sum to be at most δ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{n(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\varepsilon}^2}{n t^2 D^2} \le \delta. \tag{20}$$

Step 3 (Optimise D, then solve for t). Write $D = \eta\,|\mu_{ZX}|$ with $\eta \in (0,1)$. Dividing (20) through by δ and substituting $\rho = \sigma_{ZX}^2/(\delta n \mu_{ZX}^2)$:

$$\frac{\rho}{(1-\eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2 \eta^2 t^2} \le 1.$$

Solving for t^2:

$$t^2 \ge \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2} \cdot \frac{1}{f(\eta)}, \qquad f(\eta) = \eta^2\left(1 - \frac{\rho}{(1-\eta)^2}\right).$$

Minimising t is equivalent to maximising f over $\eta \in (0,1)$. Differentiating and setting $f'(\eta) = 0$:

$$f'(\eta) = 2\eta\left(1 - \frac{\rho}{(1-\eta)^2}\right) - \frac{2\rho\eta^2}{(1-\eta)^3} = 0.$$

Dividing through by $2\eta > 0$:

$$1 - \frac{\rho}{(1-\eta)^2} - \frac{\rho\eta}{(1-\eta)^3} = 0 \iff (1-\eta)^3 - \rho(1-\eta) - \rho\eta = 0 \iff (1-\eta)^3 = \rho.$$

Hence $\eta^* = 1 - \rho^{1/3}$, which lies in $(0,1)$ if and only if $\rho < 1$, the stated instrument strength condition. Substituting $(1-\eta^*)^2 = \rho^{2/3}$:

$$f(\eta^*) = (1-\rho^{1/3})^2\left(1 - \frac{\rho}{\rho^{2/3}}\right) = (1-\rho^{1/3})^2(1-\rho^{1/3}) = (1-\rho^{1/3})^3.$$

The minimised threshold is therefore

$$t^* = \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1-\rho^{1/3})^{3/2}},$$

and the union bound gives $\mathbb{P}[\left|\hat{\beta}_{IV} - \beta\right| > t^*] \le \delta$. □

Remark 4.6 (Comparison with the symmetric bound)**.** *As $\rho \to 0$ (strong instruments or large n), the correction factor $(1-\rho^{1/3})^{-3/2} \to 1$ and (19) approaches $\sigma_{Z\varepsilon}/(|\mu_{ZX}|\sqrt{\delta n})$, a factor of $2\sqrt{2} \approx 2.83$ tighter than (8). As $\rho \to 1$ (approaching the boundary of instrument relevance), the factor diverges, reflecting the genuine difficulty of ratio estimation near instrument irrelevance. The instrument strength condition is eight times weaker than Theorem 2.3: $\rho < 1$ versus $\rho < 1/8$.*

4.2 Cantelli Improvement

The Chebyshev bound applied to $A(D)^c$ in the proofs above is two-sided. However, $A(D)^c = \{\left|\bar{S}_{ZX}\right| < D\}$ is a one-sided failure event: only a *downward* deviation of $\bar{S}_{ZX}$ from μ_{ZX} causes the denominator to be too small; an upward deviation makes it larger and is harmless. Cantelli's inequality exploits this one-sidedness.

Lemma 4.7 (Cantelli's inequality)**.** *Let W be a random variable with $\mathbb{E}[W] = \mu$ and $\mathrm{Var}(W) = \sigma^2 < \infty$. For any $\tau > 0$,*

$$\mathbb{P}[W - \mu \le -\tau] \le \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

This is strictly smaller than the Chebyshev bound σ^2/τ^2 for all $\tau > 0$, with the greatest relative gain when $\tau \approx \sigma$.

Theorem 4.8 (Standard IV concentration, Cantelli improvement)**.** *Assume (A1)–(A3) and $\mu_{ZX} > 0$. Write $\gamma_{\mathrm{IV}} = \sigma^2_{ZX}/(n\mu^2_{ZX})$ and define*

$$h(\eta) = \eta^2 \frac{\delta\big(\gamma_{\mathrm{IV}} + (1-\eta)^2\big) - \gamma_{\mathrm{IV}}}{\gamma_{\mathrm{IV}} + (1-\eta)^2}, \quad \eta \in (0,1). \tag{21}$$

If

$$n > \frac{(1-\delta)\sigma^2_{ZX}}{\delta\mu^2_{ZX}}, \quad \textit{equivalently,} \quad \gamma_{\mathrm{IV}} < \frac{\delta}{1-\delta}, \tag{22}$$

then $h(\eta) > 0$ on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{nh(\eta^*_{\mathrm{IV}})}}\right] \le \delta, \tag{23}$$

where η^*_{IV} is the maximiser of h over its feasible domain.

Proof. We use the same two-event framework with $D = \eta\mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). As in Theorem 2.3, define $A(D) = \{\bar{S}_{ZX} \ge D\}$ and $B(D, t) = \{\left|\bar{S}_{Z\varepsilon}\right| \le tD\}$. On $A(D) \cap B(D, t)$, $\left|\hat{\beta}_{IV} - \beta\right| \le t$.

Step 2 (Cantelli on A^c, Chebyshev on B^c).

Since $\mu_{ZX} > 0$ and $D < \mu_{ZX}$, the failure event satisfies

$$A(D)^c = \{\bar{S}_{ZX} < D\} \subseteq \{\bar{S}_{ZX} - \mu_{ZX} < -(\mu_{ZX} - D)\}.$$

This is a one-sided downward deviation, so Lemma 4.7 applies with $W = \bar{S}_{ZX}$, $\sigma^2 = \sigma^2_{ZX}/n$, and $\tau = \mu_{ZX} - D$:

$$\mathbb{P}[A(D)^c] \le \frac{\sigma^2_{ZX}/n}{\sigma^2_{ZX}/n + (\mu_{ZX} - D)^2}.$$

The event $B(D, t)^c = \{\left|\bar{S}_{Z\varepsilon}\right| > tD\}$ is two-sided (a large residual of either sign causes failure), so Chebyshev is appropriate:

$$\mathbb{P}[B(D, t)^c] \le \frac{\sigma^2_{Z\varepsilon}/n}{t^2 D^2}.$$

Step 3 (Constraint and optimization). Setting the total failure probability to at most δ and substituting $D = \eta\mu_{ZX}$, $\gamma_{\mathrm{IV}} = \sigma^2_{ZX}/(n\mu^2_{ZX})$:

$$\frac{\gamma_{\mathrm{IV}}}{\gamma_{\mathrm{IV}} + (1 - \eta)^2} + \frac{\sigma^2_{Z\varepsilon}}{nt^2\eta^2\mu^2_{ZX}} \le \delta.$$

Isolating the second term and solving for t^2:

$$t^2 \ge \frac{\sigma^2_{Z\varepsilon}}{n\mu^2_{ZX} h(\eta)},$$

where $h(\eta)$ is defined in (21). The function $h(\eta)$ is positive only when $\delta(\gamma_{\mathrm{IV}} + (1 - \eta)^2) > \gamma_{\mathrm{IV}}$, i.e., $(1 - \eta)^2 > \gamma_{\mathrm{IV}}(1 - \delta)/\delta$. The feasible region is non-empty if and only if $\gamma_{\mathrm{IV}} < \delta/(1 - \delta)$, which is condition (22). Minimizing t over the feasible region by maximizing h and applying the union bound gives (23). □

Remark 4.9 (When Cantelli helps)**.** *Compared with the optimised Chebyshev condition ($n > \sigma^2_{ZX}/(\delta\mu^2_{ZX})$), the Cantelli condition* (22) *is weaker by a factor of* $1/(1 - \delta)$*. At* $\delta = 0.05$ *this ratio is* $1/0.95 \approx 1.05$*, so the gain is negligible for typical significance levels. The improvement is more meaningful at moderate confidence levels: at* $\delta = 0.5$ *the factor reaches* 2*. For the standard IV estimator the Cantelli improvement is therefore primarily of theoretical interest. By contrast, for the MoR estimator (Section* **??***), Cantelli extends the valid range of the instrument strength parameter from* $\gamma < 1/4$ *to* $\gamma < 1/3$*, a qualitatively more substantial gain.*

Remark 4.10 (No closed form for η^*_{IV}). *The first-order condition $h'(\eta) = 0$ reduces to a quartic in η with no general closed-form solution. The optimised Chebyshev version (Theorem 4.5) yielded the clean form $\eta^* = 1 - \rho^{1/3}$ because the Chebyshev bound σ^2/τ^2 produces a simpler algebraic structure than the Cantelli bound $\sigma^2/(\sigma^2 + \tau^2)$. The bound* (23) *must therefore be evaluated numerically for specific parameter values.*

References

Devroye, L., Lerasle, M., Lugosi, G. & Oliveira, R. I. (2016). Sub-gaussian mean estimators.

Hoeffding, W. (1963). Probability inequalities for sums of bounded random variables. *Journal of the American statistical association*, *58*(301), 13–30.

Lugosi, G. & Mendelson, S. (2019). Mean estimation and regression under heavy-tailed distributions: A survey. *Foundations of Computational Mathematics*, *19*(5), 1145–1190.